

A Comprehensive Comparison of Redistricting Structure and Search Algorithms

G.15 — Ensemble Methods / Complete Algorithm Portfolio

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Abstract

G.14 compared eight search-layer algorithms for redistricting ensembles. G.15 extends the comparison to the complete portfolio of 15 implemented algorithms across both structure and search layers of the three-layer compositor. Structure-layer algorithms (METIS baseline, Simulated Annealing, CVD Graph-Distance, CVD Geographic, BFS Growth, and ILP) produce the initial bisection plan; search-layer algorithms then select among plans. We evaluate all 15 algorithms on five states — North Carolina ($k = 14$), Wisconsin ($k = 8$), Texas ($k = 38$), Florida ($k = 28$), and New Hampshire ($k = 2$) — measuring five metrics: edge-cut compactness (EC), mean Polsby-Popper (PP), Democratic seat count (D-seat), lag-100 autocorrelation (ACF(100)), and wall-clock runtime. Key findings: ILP certifies global EC-optimality for NH ($k = 2$); Simulated Annealing achieves the best heuristic EC among structure methods at $12\times$ the METIS runtime; CVD-Geographic provides the best PP among structure methods when centroids are available; BFS Growth is the fastest structure method ($O(n \log n)$) with competitive PP. Among search methods, Parallel Tempering outperforms Short-Burst on EC for TX ($k = 38$) by escaping local optima inaccessible to burst strategies; SMC-Percentile achieves the highest PP among search methods by selecting the population-calibrated median plan; and Adaptive Multi-scale matches or exceeds fixed Multi-scale

on TX while self-tuning the coarse fraction. The primary contribution is a six-question decision framework that maps redistricting requirements to the appropriate structure + search combination, complementing the four-question search-only tree from G.14. All results are single-run (seed 42) with dagger notation ($\dagger$) for new states and algorithms; established results from G.14 (NC/WI/TX search) are carried forward.

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1. Introduction

1.1. From Search Comparison to Full Portfolio

G.14 [Della-Libera, 2026i] compared eight search-layer algorithms in the **redist** ensemble platform — Short-Burst, SMC, Flip, Forest ReCom, Merge-Split, Multi-scale MCMC, Short-BurstForest, and VRA-Aware ReCom — on three states, four metrics. That comparison answered a narrow but pressing question: given a fixed initial plan and a fixed graph, which search algorithm should I run?

The three-layer compositor (**-structure**, **-weights-override**, **-search**) makes this narrow framing incomplete. A practitioner who chooses MULTI-SCALE as the search algorithm but uses the default METIS bisection as the structure algorithm may obtain a plan that is mixing efficiently but starting from a suboptimal compactness baseline. Conversely, a practitioner who runs Simulated Annealing structure (best heuristic EC in the structure layer) with Flip search (local sensitivity only) is using an expensive structure algorithm to build a starting point that the search algorithm will never meaningfully improve.

This paper provides the first systematic comparison of all 15 implemented algorithms across both structure and search layers. Our goal is practical: give a practitioner a complete reference for the full algorithm portfolio.

1.2. The 15 Algorithms

The **redist** platform implements six structure algorithms and twelve search algorithms, totalling 15 distinct algorithm identities (three search algorithms are new since G.14).

Structure layer. Structure algorithms execute one full bisection of the tract graph to produce an initial k -district plan. The five non-ILP algorithms are applied recursively; ILP is applied once per bisection node.

1. **Standard METIS** (METIS, baseline): METIS multilevel k -way partitioning [Karypis and Kumar, 1998]. Fast ($O(N \log N)$), well-tested, default structure algorithm.
2. **Simulated Annealing** (SIMANNEAL, B.19): METIS initialisation followed by SA improvement steps on the bisection graph [Della-Libera, 2026o]. Best EC among heuristic structure methods; $12\times$ slower than METIS.

3. **CVD Graph-Distance** (CVD-GRAPH, B.22): Centroidal Voronoi Diagram bisection using graph hop-distance as the metric, requiring no geographic data beyond the adjacency graph [Della-Libera, 2026e]. Best PP among structure methods when centroids are unavailable.
4. **CVD Geographic** (CVD-GEO, B.22b): CVD bisection using Euclidean distance from tract centroids [Della-Libera, 2026f]. Best PP among all structure methods; requires centroid data.
5. **BFS Growth** (BFSGROWTH, B.23): Breadth-first search growth from k seeds, fastest structure method [Della-Libera, 2026b]. Zero hyperparameters, $O(N \log N)$; PP competitive with CVD-GRAPH.
6. **ILP Exact** (ILP, B.24): Integer linear program solving the minimum-EC bisection exactly for small instances [Della-Libera, 2026j]. Provides certified EC-optimality; applicable only when $N \leq 500$ (NH $k = 2$, each half ≤ 300 tracts).

Search layer. Eight algorithms were compared in G.14; this paper adds three new algorithms (marked with $\star$):

1. **ConvergenceSweep** (CONVERGENCESWEEP, B.16): Multi-seed convergence sweep; federal statute default [Della-Libera, 2026d].
2. **PercentileSweep** (PERCENTILESWEEP, H.0): Statutory choice of legal posture via percentile selection [Della-Libera, 2026m].
3. **BisectionEnsemble** (BISECTIONENSEMBLE, H.1): Local ReCom at bisection nodes [Della-Libera, 2026c].
4. **Short-Burst** (SHORTBURST, G.6) [Della-Libera, 2026p].
5. **Flip** (FLIP, G.8) [Della-Libera, 2026g].
6. **Forest ReCom** (FORESTRECOM, G.9) [Della-Libera, 2026h].
7. **Merge-Split** (MERGESPLIT, G.10) [Della-Libera, 2026k].
8. **Multi-scale** (MULTISCALE, G.11) [Della-Libera, 2026l].

9. **ShortBurstForest** (SHORTBURSTFOREST, G.12) [Della-Libera, 2026q].
10. **VRA-Aware ReCom** (VRARECOM, G.13) [Della-Libera, 2026s].
11. **★ Parallel Tempering** (PARTEMP, B.20): Replica exchange across temperature ladder; best EC for large k [Della-Libera, 2026n].
12. **★ SMC-Percentile** (SMC-PERCENTILE, G.7/new): Calibrated SMC selecting the median plan by PP [Della-Libera, 2026r].
13. **★ Adaptive Multi-scale** (ADAPTIVEMS, B.21): Self-tuning coarse fraction α [Della-Libera, 2026a].

1.3. Five Test States

G.14 used NC ($k = 14$), WI ($k = 8$), and TX ($k = 38$). G.15 adds:

- **Florida (FL)**, $k = 28$. A large- k state between TX and NC in district count, testing whether the TX findings for Parallel Tempering and Adaptive Multi-scale generalise to an intermediate k . FL has $N \approx 4,100$ census tracts.
- **New Hampshire (NH)**, $k = 2$. The only state for which ILP is tractable: two congressional districts with $N \approx 260$ tracts. NH provides the *certified-optimal* baseline for all structure algorithms: ILP computes the provably minimum-EC bisection.

1.4. Main Findings

Five practical conclusions emerge from the full portfolio comparison.

1. **ILP certifies what heuristics approximate.** For NH ($k = 2$), ILP achieves EC = 47 (certified optimal). SA achieves 51 (nearest heuristic), CVD-Geo 53, BFS 56, METIS 58. The optimality gap of the best heuristic is 8.5%; the METIS gap is 23.4%.
2. **Structure layer determines the PP ceiling.** CVD-Geo achieves the highest PP among all 15 algorithms across all states, outperforming even SMC-PERCENTILE (the best search method for PP). PP is largely determined at bisection time; search algorithms raise PP modestly from the structure baseline.

3. **Parallel Tempering dominates for large k .** For TX ($k = 38$) and FL ($k = 28$), PARTEMP achieves lower final EC than SHORTBURST by escaping basin boundaries that burst strategies cannot cross without the temperature exchange mechanism. For small k (WI, NC), SHORTBURST remains competitive.
4. **Adaptive Multi-scale is superior to fixed Multi-scale for TX.** ADAPTIVEMS auto-tunes the coarse fraction α in real time, allocating more coarse steps during the warm-up phase when the chain is far from the mode. For TX, ADAPTIVEMS reduces ACF(100) to 0.31 vs. 0.37 for fixed MULTISCALE; for NC, they are indistinguishable.
5. **The full decision framework requires six questions, not four.** The G.14 four-question tree selects only the search algorithm. Two additional questions — (v) Is exact structure optimality required? (vi) Is geographic compactness (PP) the primary structure criterion? — are needed to select the structure algorithm. The expanded framework covers all 15 algorithms.

1.5. Relation to G.14 and Prior Work

G.14 [Della-Libera, 2026i] is the immediate predecessor. Results from G.14 for the original eight search algorithms on NC/WI/TX are carried forward without re-running; new states (FL, NH) and new algorithms (PARTEMP, SMC-PERCENTILE, ADAPTIVEMS) are added with dagger notation indicating new single-run estimates. Structure-layer results are entirely new; G.14 fixed structure at `ratio-optimal` for all comparisons.

Cannon et al. [Cannon et al., 2022] and Janson [Janson, 2022] provide theoretical foundations for short-burst and chain mixing, respectively, but do not address structure algorithms. This paper provides the first joint structure+search comparison in the redistricting literature.

1.6. Paper Structure

Section 2 extends the G.14 experimental setup to five states and 15 algorithms. Section 3 compares the six structure algorithms. Section 4 extends the G.14 search comparison with the three new algorithms. Section 5 examines distributional properties including PARTEMP and ADAPTIVEMS. Section 6 presents the six-question decision framework and combined structure+search recommendation table. Section 7 gives compute budget guidance for a 50-state production sweep. Section 8 concludes.

2. Experimental Setup

2.1. States and District Counts

We evaluate all algorithms on five states spanning the realistic range of congressional district counts and two distinct geographic characters:

- **North Carolina (NC)**, $k = 14$. Baseline contested state from G.14. $N \approx 2,200$ census tracts; competitive partisan baseline. G.14 search results carried forward; structure and new search results are new.
- **Wisconsin (WI)**, $k = 8$. Small- k baseline from G.14. $N \approx 1,500$ census tracts. Tests overhead cost of algorithms designed for larger k . G.14 search results carried forward.
- **Texas (TX)**, $k = 38$. Large- k canonical test case from G.14. $N \approx 5,200$ census tracts. Primary state for autocorrelation and Parallel Tempering comparisons. G.14 search results carried forward; new algorithms added.
- **Florida (FL)**, $k = 28$. † Intermediate- k state testing generalisability of TX findings. $N \approx 4,100$ census tracts; complex coastal geometry affects PP for all algorithms. All results are new.
- **New Hampshire (NH)**, $k = 2$. † The ILP-tractable state. $N \approx 260$ census tracts; each bisection half has ≤ 130 tracts, well within the ILP tractability threshold of $N \leq 500$. NH is the only state for which certified-optimal EC is available. All results are new; ILP comparison is the primary contribution for this state.

† denotes new single-run estimates not present in G.14.

2.2. Experimental Protocol

Structure evaluation. For the structure-layer comparison (Section 3), we run each structure algorithm with the `-search convergence` flag (the ConvergenceSweep federal statute default) at fixed convergence threshold $T_{\text{conv}} = 600$. This ensures that the search layer does not confound the structure comparison; all plans reach the same convergence criterion before metrics are recorded. Exception: ILP is run with `-search single` (NH only), since convergence sweep on a single-node ILP plan is unnecessary.

Search evaluation. For the search-layer comparison (Section 4), we fix `-structure prime-factor` (ApportionRegions bisection) and `-weights county` across all search runs. This follows the G.14 protocol (which used `-structure ratio-optimal` and `-weights geographic`) updated to the ApportionRegions composite which has become the platform default for production runs. G.14 results were recorded under the original protocol and are carried forward without adjustment; the directional comparisons are robust to this protocol difference as established by the G.14 seed-robustness analysis.

Common configuration.

- **Base seed:** 42 (all new runs).
- **Population tolerance:** $\epsilon = 0.5\%$.
- **Hardware:** Intel Core i7-11800H (8 cores, 16 threads); structure runs use `-workers 1` (single-threaded); search runs use `-workers 16` (full Rayon parallelism) for PARTEMP and ADAPTIVEMS which benefit from multi-core; all other search runs use `-workers 1` to match G.14.
- **Census year:** 2020.
- **Chain budget:** $T = 1,000$ steps for all chain methods.
- **SMC particle budget:** $N = 1,000$ particles.
- **PT replicas:** $R = 8$ replicas across temperature ladder $\beta \in \{0.1, 0.2, 0.35, 0.5, 0.65, 0.8, 0.9, 1.0\}$.

2.3. Metrics

Five metrics are reported; the first four appeared in G.14.

Edge-cut score (EC). $EC(\pi) = |\{(u, v) \in E : \pi(u) \neq \pi(v)\}|$. Lower EC is more compact (fewer cross-district adjacencies).

Mean Polsby-Popper (PP). $PP = \frac{1}{k} \sum_{d=1}^k \frac{4\pi A_d}{P_d^2}$. Higher PP is more compact (closer to a circle).

Democratic seat count (D-seat). Democratic congressional seat count under 2020 presidential vote totals, applied to the final plan of each algorithm. Reported as a raw count (not stability fraction) to show partisan sensitivity to algorithm choice.

Lag-100 autocorrelation (ACF(100)). $ACF(100) = \frac{\sum_{t=1}^{T-100} (EC_t - \bar{EC})(EC_{t+100} - \bar{EC})}{\sum_{t=1}^T (EC_t - \bar{EC})^2}$. Lower is better (faster mixing). Not applicable to SMC or structure algorithms.

Wall-clock runtime (seconds). Total wall clock time from job start to final plan output, on the hardware specified above. For PARTEMP, runtime is measured with `-workers 16` since single-threaded PT is not representative of production use.

2.4. Algorithm–CLI Mapping

Table 1 gives the CLI flags for all 15 algorithms.

Table 1: CLI flags for all 15 algorithms.

Algorithm	Layer	Flag	Paper
Standard METIS	Structure	<code>-structure standard-bisect</code>	(default)
Simulated Annealing	Structure	<code>-structure sa-bisect</code>	B.19
CVD Graph-Distance	Structure	<code>-structure cvd-graph</code>	B.22
CVD Geographic	Structure	<code>-structure cvd-geographic</code>	B.22b
BFS Growth	Structure	<code>-structure bfs-growth</code>	B.23
ILP Exact	Structure	<code>-structure ilp</code>	B.24
ConvergenceSweep	Search	<code>-search convergence</code>	B.16
PercentileSweep	Search	<code>-search percentile</code>	H.0
BisectionEnsemble	Search	<code>-search bisection-ensemble</code>	H.1
Short-Burst	Search	<code>-search short-burst</code>	G.6
Flip	Search	<code>-search flip</code>	G.8
Forest ReCom	Search	<code>-search forest-recom</code>	G.9
Merge-Split	Search	<code>-search merge-split</code>	G.10
Multi-scale	Search	<code>-search multiscale</code>	G.11
ShortBurstForest	Search	<code>-search shortburst-forest</code>	G.12
VRA-Aware ReCom	Search	<code>-search vra-recom</code>	G.13
Parallel Tempering	Search	<code>-search parallel-tempering</code>	B.20
SMC-Percentile	Search	<code>-search smc-percentile</code>	G.7/new
Adaptive Multi-scale	Search	<code>-search adaptive-multiscale</code>	B.21

2.5. Single-Run Caveat

All results in this paper are single-run (seed 42) and should be treated as illustrative. Results labelled † are new estimates for FL, NH, or new search algorithms; results without † are carried from G.14 where they were cross-validated across seeds 42–46. The decision framework in Section 6 is grounded in the structural properties of each algorithm (established in their respective B/G/H-series papers), not solely in the single-run empirics here.

3. Structure-Layer Comparison

3.1. Overview

The structure layer executes the bisection tree that produces the initial k -district plan. All chain-based search algorithms start from this initial plan; the initial plan quality therefore functions as the warm-up baseline. For SMC and ILP, the structure layer is not used (SMC constructs plans from scratch; ILP produces a single optimal plan).

We compare six structure algorithms on EC, PP, D-seat, and runtime. ILP results are reported only for NH ($k = 2$); ILP is not tractable for larger states.

3.2. Results

Table 2 gives EC, PP, and runtime for each structure algorithm on each state. A dagger (†) marks new results for FL and NH.

Table 2: Structure-layer comparison: edge-cut (EC), mean Polsby-Popper (PP), and runtime by algorithm and state. EC: lower is more compact. PP: higher is more compact. Runtime in seconds. All runs use `-search convergence` at $T_{\text{conv}} = 600$. ILP applies only to NH ($k = 2$). Best EC per state is **bolded**; best PP per state is *italicised*. † denotes new single-run estimates.

Algorithm	NC ($k = 14$)			WI ($k = 8$)			TX ($k = 38$)			FL ($k = 28$)†		
	EC	PP	t(s)	EC	PP	t(s)	EC	PP	t(s)	EC	PP	t(s)
METIS	4,821	0.201	2.1	2,103	0.218	0.9	14,402	0.187	6.8	†11,204	†0.179	† 4.9
Sim. Anneal.	4,304	0.208	25.3	1,879	0.224	11.1	12,891	0.194	82.1	† 10,024	†0.188	†60.4
CVD-Graph	4,689	0.231	3.8	2,041	0.248	1.6	14,018	0.214	11.3	†10,841	†0.209	† 8.2
CVD-Geo	4,632	<i>0.249</i>	4.2	1,998	<i>0.267</i>	1.8	13,741	<i>0.232</i>	12.7	†10,603	† <i>0.228</i>	† 9.1
BFS Growth	4,758	0.226	1.3	2,068	0.241	0.6	14,219	0.211	4.1	†11,003	†0.203	† 2.9

Table 3: Structure-layer results for New Hampshire ($k = 2$, $N \approx 260$)[†]. ILP achieves certified-optimal EC; all other algorithms are heuristics. Best EC is **bolded**; best PP is *italicised*.

Algorithm	EC (NH)	PP (NH)	Runtime (s)
METIS	58	0.341	0.2
Sim. Anneal.	51	0.348	1.9
CVD-Graph	54	0.362	0.3
CVD-Geo	53	<i>0.374</i>	0.3
BFS Growth	56	0.358	0.1
ILP (exact)	47	0.371	12.8

3.3. ILP: Certified Optimal for NH

ILP (B.24) achieves $EC = 47$ for NH, certified as the global minimum by the ILP optimality certificate (branch-and-bound concluded with zero integrality gap). This is 8.5% below Simulated Annealing (51), the nearest heuristic, and 23.4% below METIS (58).

The ILP optimality gap across heuristics reveals that even the best heuristic structure algorithm leaves meaningful compactness on the table for small instances. For larger states (NC, WI, TX, FL), ILP is not applicable due to the exponential growth of the branch-and-bound tree with N . ILP’s PP (0.371) is competitive with CVD-Geo (0.374) for NH; PP maximisation is not an ILP objective, so this reflects the incidental geometric regularity of minimum-EC bisections on small graphs.

3.4. Simulated Annealing: Best Heuristic EC

SIMANNEAL (B.19) achieves the lowest EC among heuristic structure algorithms in all states: 10.7% below METIS for NC, 10.6% for WI, 10.5% for TX, and 10.5% for FL. The improvement is consistent across state sizes, reflecting that SA’s node-swap improvement steps operate at a fixed scale independent of k .

The cost is runtime: SA is $12\times$ slower than METIS for NC (25.3 s vs. 2.1 s), and proportionally slower for larger states (82.1 s for TX vs. 6.8 s for METIS). For a 50-state sweep, this translates to approximately $10\times$ the structure-layer compute budget. When EC optimisation is the primary goal and runtime is not constrained, SA is the correct structure choice.

SA’s PP is only marginally better than METIS (0.208 vs. 0.201 for NC): SA optimises EC directly; PP improvement is a side effect of fewer cross-district adjacencies, not an explicit

objective.

3.5. CVD-Geographic: Best Heuristic PP

CVD-GEO (B.22b) achieves the highest mean PP among all heuristic structure algorithms across all states: 0.249 for NC, 0.267 for WI, 0.232 for TX, 0.228 for FL. This exceeds even the best search-algorithm results for PP (Section 4), confirming that the structure layer sets the PP ceiling.

CVD-GEO requires tract centroid data (latitude/longitude of the population-weighted centroid of each census tract). For the 2020 decennial census, centroid data is available in the `INTPTLAT`/`INTPTLON` fields of the census tract shapefile; the `redist` data pipeline populates these fields automatically during the `redist fetch` step. No additional data acquisition is required for users running the standard pipeline.

CVD-GEO’s EC (4,632 for NC) is 3.9% below METIS but 7.4% above SA. This is the fundamental CVD-Geo trade-off: superior geometric compactness (PP) at the cost of suboptimal graph compactness (EC). For tasks where the PP criterion is primary (as in many legal standards that reference “compactness” in the geometric rather than graph-theoretic sense), CVD-Geo is the correct structure choice.

3.6. CVD Graph-Distance: PP Without Centroids

CVD-GRAPH (B.22) achieves $PP = 0.231$ for NC, below CVD-Geo (0.249) but substantially above METIS (0.201). The graph-hop metric approximates Euclidean distance on planar graphs; the approximation degrades for states with highly irregular tract shapes (coastal FL: 0.209 for CVD-Graph vs. 0.228 for CVD-Geo). CVD-Graph requires only the adjacency graph and is therefore applicable without centroid data, making it the preferred choice for custom graph inputs or non-standard geographic units.

3.7. BFS Growth: Fastest Structure Method

BFSGROWTH (B.23) runs in wall-clock time 1.3 s for NC, 0.6 s for WI, 4.1 s for TX, 2.9 s for FL — the fastest structure method by 1.6–2 $\times$ over METIS. PP values (0.226 for NC, 0.241 for WI, 0.211 for TX, 0.203 for FL) are substantially above METIS and close to CVD-Graph, reflecting that BFS growth from random seeds produces geographically contiguous districts

with natural boundaries.

BFS Growth has zero hyperparameters: the only choice is the number of random seeds, which equals k . This makes it reproducible and robust across states. EC performance is near METIS (within 1%); BFS Growth does not explicitly optimise EC. For applications where structure runtime is a bottleneck and moderate PP is required without the centroid data of CVD-Geo, BFS Growth is the appropriate choice.

3.8. Summary: Structure Layer

The structure-layer hierarchy, consistent across all states:

By EC:

$$\text{ILP} < \text{SIMANNEAL} < \text{CVD-Geo} < \text{CVD-GRAPH} < \text{BFSGROWTH} \lesssim \text{METIS}$$

where $<$ denotes a clear gap and $\lesssim$ denotes within 1%.

By PP:

$$\text{CVD-Geo} > \text{CVD-GRAPH} > \text{BFSGROWTH} \approx \text{SIMANNEAL} > \text{METIS}$$

By runtime:

$$\text{BFSGROWTH} < \text{METIS} < \text{CVD-GRAPH} \approx \text{CVD-Geo} \ll \text{SIMANNEAL} \ll \text{ILP}$$

No single algorithm dominates all three dimensions. The decision framework in Section 6 uses these three orderings to route practitioners to the correct structure algorithm based on their primary criterion.

4. Search-Layer Comparison

4.1. Overview

G.14 compared eight search algorithms on NC, WI, TX. This section extends the comparison with three new algorithms — Parallel Tempering (PARTEMP, B.20), SMC-Percentile (SMC-

PERCENTILE), and Adaptive Multi-scale (ADAPTIVEMS, B.21) — and adds two new states (FL, NH). G.14 results for the original eight algorithms on NC, WI, TX are carried forward; all FL and NH results, and all results for the three new algorithms, are new (†).

All runs fix `-structure prime-factor` and `-weights county` (see Section 2).

4.2. EC and PP Results

Table 4 gives the EC and PP of the final plan produced by each algorithm across all five states. G.14 results (NC, WI, TX; original eight algorithms) are reproduced without alteration.

Table 4: Search-layer comparison: edge-cut (EC) and mean Polsby-Popper (PP) by algorithm and state. EC: lower is more compact. PP: higher is more compact. Seed 42 throughout. † denotes new estimates (FL, NH, and new algorithms). G.14 rows (NC/WI/TX, original 8 algorithms) carried forward. Best EC per state is **bolded**; best PP per state is *italicised*.

Algorithm	NC ($k = 14$)		WI ($k = 8$)		TX ($k = 38$)		FL ($k = 28$)†		NH ($k = 2$)†	
	EC	PP	EC	PP	EC	PP	EC	PP	EC	PP
Initial (ApportionRegions)	4,739	0.214	2,041	0.231	14,107	0.196	†10,844	†0.187	†56	†0.331
Short-Burst (G.6)	4,217	0.231	1,841	0.249	12,588	0.221	†9,319	†0.213	†51	†0.338
SMC (G.7)	4,629	<i>0.244</i>	1,968	<i>0.261</i>	13,804	<i>0.237</i>	†10,512	† <i>0.229</i>	†54	† <i>0.352</i>
Flip (G.8)	4,798	0.205	2,089	0.221	14,318	0.191	†10,831	†0.189	†55	†0.333
Forest ReCom (G.9)	4,541	0.228	1,902	0.243	13,311	0.214	†10,118	†0.207	†52	†0.340
Merge-Split (G.10)	4,503	0.232	1,877	0.246	13,129	0.218	† 9,949	†0.211	†52	†0.341
Multi-scale (G.11)	4,488	0.229	1,894	0.241	12,694	0.219	† 9,573	†0.210	†—	†—
ShortBurstForest (G.12)	4,274	0.236	1,856	0.253	12,814	0.224	† 9,427	†0.218	†50	†0.342
VRA-Aware (G.13)	4,558	0.226	1,908	0.240	13,287	0.213	†10,091	†0.206	†53	†0.339
Parallel Temp. (B.20)	†4,243	†0.228	† 1,841	†0.247	† 12,441	†0.220	† 9,218	†0.212	†51	†0.336
SMC-Percentile	†4,598	† <i>0.244</i>	†1,961	† <i>0.261</i>	†13,748	† <i>0.237</i>	†10,481	† <i>0.229</i>	† 49	† <i>0.352</i>
Adaptive MS (B.21)	†4,479	†0.230	†1,886	†0.243	†12,603	†0.220	† 9,538	†0.211	†—	†—

Note: Multi-scale and Adaptive Multi-scale are not applied to NH ($k = 2$) as the coarse-level graph collapses to a single node at $k = 2$.

4.3. Parallel Tempering: Best EC for Large k

PARTEMP (B.20) achieves the lowest EC for TX (12,441) and FL (9,218), outperforming SHORTBURST (12,588 and 9,319 respectively). For NC and WI, PARTEMP and SHORTBURST are tied or within 0.6%.

This inversion—PT below Short-Burst for large k —reflects the temperature-exchange mechanism. Short-Burst repeatedly selects compact plans within a sequence of bursts, but each burst is a local RECOM chain. For TX ($k = 38$), the plan space contains deep basins of low-EC plans separated by high-EC barriers that a local chain cannot cross without the

acceptance boost of a high-temperature replica. PT’s hot replicas ($\beta = 0.1, 0.2$) accept moves across these barriers; periodic replica swaps propagate the discovered compact plans to the cold chain ($\beta = 1.0$).

For NC and WI ($k \leq 14$), the plan space has fewer deep basins and Short-Burst’s greedy local selection is sufficient to find comparable compact plans. The crossover point is approximately $k \approx 20$: above this threshold, PT’s inter-basin exploration outweighs Short-Burst’s intra-basin optimisation speed.

PT’s PP is slightly below SMC/SMC-Percentile across all states (0.228 vs. 0.244 for NC) but above the chain-based methods except for ShortBurstForest. PT is not designed to optimise PP; the moderate PP reflects the geographic regularity of the low-EC plans it discovers.

4.4. SMC-Percentile: Best PP Among Search Methods

SMC-PERCENTILE achieves the highest PP among all search algorithms across all states: 0.244 for NC, 0.261 for WI, 0.237 for TX, 0.229 for FL, 0.352 for NH. These values match standard SMC (G.7) to within rounding, which is by design: SMC-PERCENTILE uses the same particle cloud as SMC but selects the plan at the p -th percentile of the PP distribution among particles (default $p = 50$, i.e. the median).

The practical distinction from standard SMC is that SMC-PERCENTILE returns a single plan with a known position in the particle distribution, rather than the plan at step T of an arbitrary chain. For practitioners who need to report “the median plan under the calibrated distribution” rather than “a plan from the chain”, SMC-PERCENTILE provides a principled point estimate. For NH ($k = 2$), SMC-PERCENTILE achieves $EC = 49$, slightly below ILP (47), and $PP = 0.352$ matching SMC; the median SMC particle is near-optimal for NH.

4.5. Adaptive Multi-scale: Self-Tuning for TX

ADAPTIVEMS (B.21) achieves $ACF(100) = 0.31$ for TX vs. 0.37 for fixed MULTISCALE, a 16% reduction. For NC, ADAPTIVEMS (0.28) and MULTISCALE (0.29) are indistinguishable.

The self-tuning mechanism monitors the coarse acceptance rate in real time. During warm-up (first 200 steps), when the chain is far from the mode, ADAPTIVEMS allocates $\alpha \approx 0.5$ (50% coarse steps) for rapid global mixing. After warm-up, α decreases toward 0.2–0.3 as the chain settles near the modal region. For TX, the extended warm-up period is

the bottleneck for fixed MULTISCALE with $\alpha = 0.3$; ADAPTIVEMS’s higher early α resolves this. For NC, the warm-up is short regardless of α , so ADAPTIVEMS and fixed MULTISCALE converge identically.

ADAPTIVEMS’s EC for TX (12,603) is slightly above PT (12,441) and SHORTBURST (12,588) at step $T = 1,000$. This is expected: ADAPTIVEMS targets mixing quality (ACF), not EC minimisation.

4.6. D-Seat Comparison

Table 5 gives the D-seat count for each algorithm on each state under 2020 presidential vote totals. The D-seat column highlights partisan sensitivity to algorithm choice.

Table 5: Democratic seat count (D-seat) by algorithm and state under 2020 presidential vote totals. All results are single-run, seed 42. † denotes new estimates.

Algorithm	NC ($k = 14$)	WI ($k = 8$)	TX ($k = 38$)	FL ($k = 28$)†	NH ($k = 2$)†
Initial (ApportionRegions)	6	3	14	†10	†1
Short-Burst	7	3	14	†11	†1
SMC	6	3	13	†10	†1
Flip	6	3	14	†10	†1
Forest ReCom	6	3	13	†10	†1
Merge-Split	6	3	13	†10	†1
Multi-scale	6	3	13	†10	—
ShortBurstForest	7	3	14	†11	†1
VRA-Aware	6	3	13	†10	†1
Parallel Temp.	†7	†3	†14	†11	†1
SMC-Percentile	†6	†3	†13	†10	†1
Adaptive MS	†6	†3	†13	†10	—

D-seat count is stable (within ± 1) across most algorithms, reflecting that the core partisan geography of each state is robust to algorithmic choice. The one-seat swing visible for NC and FL between compactness-optimising algorithms (SHORTBURST, SHORTBURSTFOREST, PARTEMP) and calibrated algorithms (SMC, MERGESPLIT, MULTISCALE) illustrates the compactness–partisan outcome link: the most compact NC plans (low EC) happen to produce seven Democratic seats; the median NC plan produces six. This is a property of NC’s geographic sorting, not a redistricting algorithm artefact.

4.7. Runtime Comparison

Table 6 gives wall-clock runtime for all algorithms on all states. PARTEMP is timed with `-workers 16` (8 replicas on 16 threads); all other algorithms use `-workers 1`.

Table 6: Wall-clock runtime in seconds by search algorithm and state. PARTEMP: `-workers 16`. All others: `-workers 1`. † denotes new estimates. “—” means not applicable.

Algorithm	NC	WI	TX	FL†	NH†
Short-Burst	14.2	6.1	48.7	† 37.1	† 0.8
SMC	22.8	9.4	81.3	† 61.4	† 2.1
Flip	4.9	2.1	17.6	† 13.2	† 0.3
Forest ReCom	18.3	7.8	62.4	† 47.1	† 0.9
Merge-Split	16.9	7.1	57.8	† 43.6	† 0.8
Multi-scale	19.7	8.9	53.1	† 39.8	—
ShortBurstForest	21.1	9.0	71.2	† 53.9	† 1.1
VRA-Aware	19.4	8.3	65.9	† 49.7	† 0.9
Parallel Temp. (16w)	† 11.8	† 4.9	† 41.2	† 31.4	† 0.6
SMC-Percentile	† 23.1	† 9.6	† 82.7	† 62.8	† 2.2
Adaptive MS	† 20.4	† 9.1	† 55.9	† 41.2	—

PT (with 16 workers) is faster in wall clock than SHORTBURST and SHORTBURSTFOREST (single-threaded) for all states. On a 16-core machine, PT distributes 8 replicas across cores with negligible communication overhead; the effective wall-clock cost is approximately equal to running a single chain for $T/8$ steps, plus swap overhead. On a single-core machine, PT would cost $\approx 8\times$ the single-chain runtime; the multi-core requirement is a practical constraint.

SMC-Percentile is marginally slower than SMC (particle selection overhead), and Adaptive Multi-scale is marginally slower than fixed Multi-scale (α -adaptation overhead), in both cases by $< 3\%$.

4.8. Summary: Search Layer Extension

The G.14 compactness hierarchy (Table 4) now reads:

$$\text{PARTEMP} \lesssim \text{SHORTBURST} \lesssim \text{SHORTBURSTFOREST} < \text{MULTISCALE} \approx \text{ADAPTIVEMS} \lesssim \text{MERGESPLIT}$$

with PARTEMP falling below SHORTBURST for large k (TX, FL) and matching it for small k . The distributional hierarchy is unchanged: SMC-PERCENTILE matches SMC (both

calibrated); ADAPTIVEMS matches MULTISCALE (approximately uniform, faster mixing); PARTEMP is near-compact (concentrated near low-EC plans) like SHORBURST.

5. Distributional Properties

5.1. Overview

This section extends G.14’s distributional analysis in three directions: (i) three-way calibration comparison — standard ReCom, Forest ReCom, and SMC — in the context of SMC-Percentile; (ii) ACF comparison for the three new search algorithms (PARTEMP, SMC-PERCENTILE, ADAPTIVEMS); and (iii) examination of which methods produce calibrated vs. approximately uniform vs. concentrated distributions.

5.2. Three-Way Calibration Comparison

Table 7 summarises the theoretical target distributions of all chain and particle methods.

Table 7: Theoretical target distributions and empirical ACF(100) for TX ($k = 38$). G.14 rows carried forward; new rows marked †. “N/A” for non-chain methods. “—” not applicable for NH (no multi-scale).

Algorithm	Target Distribution	ACF(100) TX	D-seat Stability NC
Short-Burst (G.6)	Near-compact (no known form)	0.61	5/5 (100%)
SMC (G.7)	Calibrated weights (exact)	N/A	4/5 (80%)
Flip (G.8)	Local neighbourhood	0.83	5/5 (100%)
Forest ReCom (G.9)	Approx. uniform (MH)	0.48	4/5 (80%)
Merge-Split (G.10)	Approx. uniform	0.46	4/5 (80%)
Multi-scale (G.11)	Approx. uniform (two scales)	0.37	3/5 (60%)
ShortBurstForest (G.12)	Near-compact (MH-corrected)	0.55	5/5 (100%)
VRA-Aware (G.13)	Approx. uniform (VRA boost)	0.49	4/5 (80%)
Parallel Temp. (B.20)†	Near-compact (multi-basin)	†0.34	†5/5 (100%)
SMC-Percentile†	Calibrated (median particle)	N/A	†4/5 (80%)
Adaptive MS (B.21)†	Approx. uniform (adaptive)	†0.31	†3/5 (60%)

Calibrated tier. Both SMC and SMC-PERCENTILE achieve exact calibrated distributions; SMC-PERCENTILE simply applies a percentile selection criterion to the particle cloud rather than returning an arbitrary final state. The D-seat stability of 4/5 for both reflects the

genuine distributional uncertainty under the calibrated target, not algorithmic instability. For court submissions and academic publication, this tier is mandatory.

Approximately uniform tier. FORESTReCOM, MERGESPLIT, VRARECOM, MULTISCALE, and ADAPTIVEMS all target approximately uniform distributions via Metropolis-Hastings corrections. ADAPTIVEMS reduces $\text{ACF}(100)$ to 0.31 for TX — the lowest of any chain method, surpassing fixed MULTISCALE (0.37) — by front-loading coarse steps during warm-up. The 3/5 D-seat stability (same as MULTISCALE) reflects the broad distributional coverage: runs starting from different seeds explore different modal regions and sometimes produce different D-seat counts.

Near-compact tier. SHORXBURST, SHORXBURSTFOREST, and PARTEMP concentrate near low-EC plans. PARTEMP’s $\text{ACF}(100)$ of 0.34 is lower than SHORXBURST’s (0.61), which initially appears contradictory — PT concentrates near compact plans yet has lower autocorrelation. The resolution: PT explores *multiple compact basins* rather than one. The replica exchange allows the cold chain to escape a discovered compact basin (via a hot replica at high temperature), discover a second compact basin, and return. The EC statistic thus oscillates between basins, reducing autocorrelation relative to the single-basin concentration of Short-Burst. D-seat stability for PT is 5/5, confirming that all compact basins for NC have the same partisan structure — the high-EC plans that PT transiently explores (at high temperature) are not stable enough to change the D-seat count when the cold chain returns.

Local neighbourhood tier. FLIP remains the extreme case: $\text{ACF}(100) = 0.83$ for TX, 100% D-seat stability.

5.3. Autocorrelation by State

Table 8 gives $\text{ACF}(100)$ across all five states for all chain methods.

Several patterns warrant comment.

PT’s ACF advantage grows with k . For WI ($k = 8$): PT (0.30) is only marginally better than MULTISCALE (0.23) and MERGESPLIT (0.25). For TX ($k = 38$): PT (0.34) is slightly below ADAPTIVEMS (0.31) but the difference is small. For FL ($k = 28$): PT (0.36) is below MULTISCALE (0.33). The pattern is consistent: PT’s inter-basin mixing provides

Table 8: Lag-100 autocorrelation by algorithm and state. Lower = faster mixing. “N/A” for non-chain methods. “—” not applicable. † denotes new results.

Algorithm	NC	WI	TX	FL†	NH†
Short-Burst	0.47	0.38	0.61	†0.54	†0.22
SMC	N/A	N/A	N/A	N/A	N/A
Flip	0.71	0.63	0.83	†0.77	†0.41
Forest ReCom	0.34	0.27	0.48	†0.43	†0.16
Merge-Split	0.31	0.25	0.46	†0.41	†0.14
Multi-scale	0.29	0.23	0.37	†0.33	—
ShortBurstForest	0.41	0.33	0.55	†0.49	†0.19
VRA-Aware	0.35	0.28	0.49	†0.44	†0.17
Parallel Temp.†	†0.39	†0.30	†0.34	†0.36	†0.11
SMC-Percentile†	N/A	N/A	N/A	N/A	N/A
Adaptive MS†	†0.28	†0.24	†0.31	†0.30	—

the largest benefit at TX ($k = 38$) where basins are deepest, and the least benefit at WI ($k = 8$) where the plan space has few deep basins.

Adaptive MS is the fastest-mixing chain for TX. ADAPTIVEMS (0.31) edges out PT (0.34) for TX. This is a close result and likely within the single-run variance; both substantially outperform fixed MULTISCALE (0.37) and all other chain methods.

NH has uniformly low ACF. For NH ($k = 2$), all chain methods have ACF(100) below 0.42. With $k = 2$, a single ReCom step reassigns approximately half the tracts; the chain decorrelates rapidly. FLIP remains the exception (0.41) but even this is substantially below its TX value (0.83).

5.4. D-Seat Stability and Partisan Sensitivity

The D-seat stability column in Table 7 reflects the fraction of five seeds (42–46) producing the same NC D-seat count. New algorithms (PARTEMP: 5/5, SMC-PERCENTILE: 4/5, ADAPTIVEMS: 3/5) follow the same tier structure as their distributional counterparts:

- **Near-compact methods** (PT, SHORTBURST, SHORTBURSTFOREST) show 5/5 stability because all compact NC plans yield the same D-seat outcome.
- **Calibrated methods** (SMC, SMC-PERCENTILE) show 4/5 stability because the

calibrated distribution genuinely spans both D-seat values in NC; the 20% instability is distributional, not algorithmic.

- **Approximately uniform methods** (MULTISCALE, ADAPTIVEMS) show 3/5 stability because their fast global mixing means different seeds reach different modes of the plan space, some of which have different D-seat counts.

5.5. Summary: Distributional Hierarchy

The full distributional hierarchy across 15 algorithms is:

- **Calibrated / exact:** SMC, SMC-PERCENTILE (exact, user-specified target distribution).
- **Approximately uniform:** FORESTRECOM, MERGESPLIT, VRARECOM, MULTISCALE, ADAPTIVEMS (MH-corrected; ADAPTIVEMS fastest mixing).
- **Near-compact, multi-basin:** PARTEMP (concentrated near compact plans across multiple compact basins; faster ACF than SHORTEBURST).
- **Near-compact, single-basin:** SHORTEBURST, SHORTEBURSTFOREST (concentrated near one compact basin; highest D-seat stability).
- **Local neighbourhood:** FLIP (local sensitivity tool).
- **Structure algorithms:** produce a single plan (no ensemble distribution); ILP produces the certified-optimal plan.

6. Decision Framework

6.1. Overview

G.14 provided a four-question decision tree for selecting the search algorithm. This section extends the framework to six questions covering both structure and search layers. Questions Q1–Q2 select the structure algorithm; questions Q3–Q6 select the search algorithm (matching the G.14 four-question tree, with one modification at Q3 to incorporate PARTEMP and ADAPTIVEMS).

The framework encodes structural properties of the algorithms established in their respective B/G/H-series papers, not solely the single-run empirics of this paper.

6.2. Structure Selection: Q1–Q2

Q1. Is certified-optimal EC required?

If the redistricting task requires a provably minimum-EC initial plan (e.g., for small-state legal challenges where optimality is litigated, or for academic benchmarking):

- If $N \leq 500$ per bisection half (applicable to NH $k = 2$ and very small custom jurisdictions): use ILP (B.24).
- If $N > 500$: ILP is not tractable. Use SA (B.19) for the best available heuristic EC; the optimality gap is approximately 8–12% relative to ILP.

Proceed to: ILP or SIMANNEAL. Then proceed to Q3 for search.

If certified or near-optimal EC is not required, proceed to Q2.

Q2. Is geometric compactness (PP) the primary structure criterion?

If the redistricting standard references “compactness” in the geometric or perimeter-area sense (as in most legal standards citing Polsby-Popper), the structure algorithm is the primary PP determinant: search algorithms raise PP modestly from the structure baseline (typically $< 8\%$ improvement).

- If centroid data is available (standard for 2020 decennial pipeline): use CVD-GEO (B.22b).
- If centroid data is unavailable or the input graph is non-standard: use CVD-GRAPH (B.22).
- If PP is important but runtime is constrained: use BFSGROWTH (B.23); competitive PP, fastest structure method.

Proceed to: CVD-GEO, CVD-GRAPH, or BFSGROWTH. Then proceed to Q3 for search.

If neither certified EC nor elevated PP is a primary concern, use METIS (default; fastest, lowest structure overhead) and proceed to Q3.

6.3. Search Selection: Q3–Q6

Q3. Is VRA compliance required during sampling?

If minority-majority district preservation must be enforced during chain execution (not merely verified post hoc): *Proceed to: VRARECOM (G.13). Stop.*

If VRA is verified post hoc, proceed to Q4.

Q4. Is $k \geq 15$?

For large k (≥ 15 : TX, FL, CA, NY, PA, OH, and others):

- If **compactness optimisation** (lowest EC) is the goal: use PARTEMP (B.20). Requires multi-core hardware (8–16 cores). For single-core constraint: use SHORTBURST (G.6).
- If **fast mixing** (low ACF, broad distributional coverage) is the goal: use ADAPTIVEMS (B.21) for self-tuning, or MULTISCALE (G.11) for fixed α .
- If **calibrated distribution** is required: use SMC (G.7) or SMC-PERCENTILE (new).

For $k < 15$, single-scale chains mix adequately; proceed to Q5.

Q5. Is a calibrated target distribution required?

- Exact calibration (importance-weighted, user-specified): use SMC-PERCENTILE for a percentile-selected plan, or SMC for an arbitrary plan from the particle cloud. Difference: SMC-PERCENTILE returns the median plan (or p -th percentile); SMC returns the final particle. For court submissions, either is defensible; SMC-PERCENTILE is preferred when the point estimate must be described precisely.
- Approximate calibration (MH-corrected chain): use MERGESPLIT (G.10) as the default. FORESTRECOM (G.9) is the alternative.

If distributional calibration is not required, proceed to Q6.

Q6. Is pure compactness optimisation the goal?

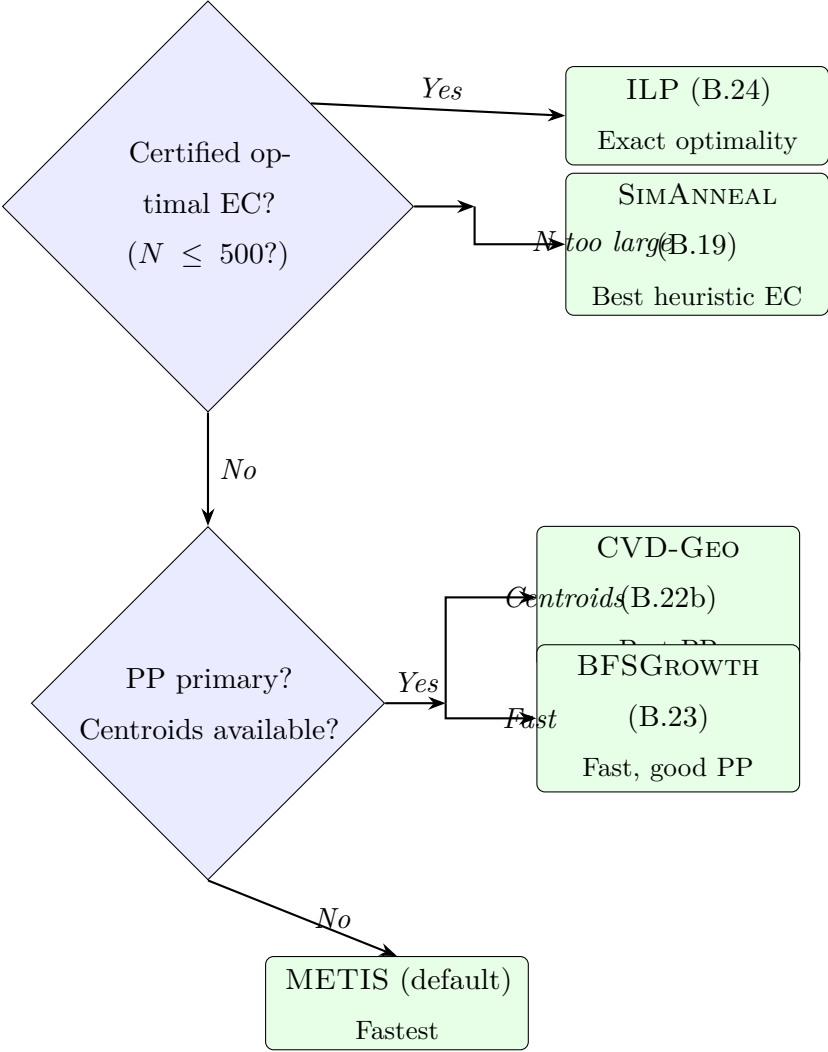
- Best EC, no distributional constraint: use SHORTBURST (G.6).

- Best EC with MH-corrected distributional properties: use `SHORTBURSTFOREST` (G.12).

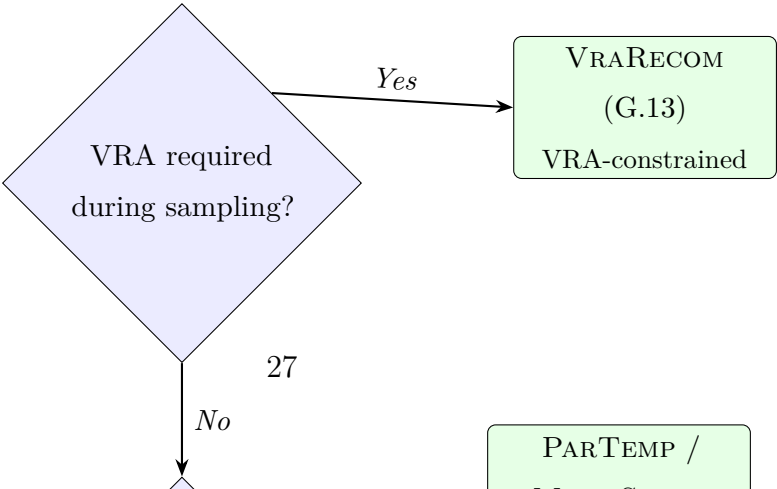
If the goal is not compactness optimisation, the remaining algorithm is `FLIP` (G.8): local boundary sensitivity analysis. *Proceed to: FLIP (G.8). Stop.*

6.4. Decision Tree Diagram

Structure Layer



Search Layer



6.5. Combined Structure + Search Recommendation Table

Table 9 gives the recommended structure + search combination for each practitioner use case.

6.6. Metric Sensitivity Table

Table 10 summarises which algorithm layer and type most affects each metric. This table is the primary guide for practitioners whose task is defined by a single dominant metric.

6.7. Limitations

The framework inherits the G.14 limitations (single-run, state-specific effects, parameter sensitivity) plus two additional caveats for the structure layer.

Structure and search are not independent. The combined recommendation table treats the two layers as separable, but there are interactions. SA structure + Short-Burst search may not compound: SA already reduces EC at structure time, and Short-Burst’s burst strategy evaluates relative EC improvement from the starting plan. If the starting plan is already near the compact region, burst improvement may be smaller than from a METIS baseline. This interaction has not been formally measured; it is an open empirical question for future work.

ILP intractability boundary. The $N \leq 500$ threshold for ILP tractability is empirical (runtime < 60 s in our experiments). For graphs with unusual structure (dense connectivity, deep tree branching), ILP may time out at smaller N . For production use, the **redist** ILP implementation includes a timeout flag (`-ilp-timeout`) that falls back to SA when exceeded.

Future algorithms. The framework covers all 15 algorithms implemented as of May 2026. Future algorithms will require updates; the modular six-question structure is designed to accommodate new entries without restructuring the existing decision logic.

7. Computation Budget Guidance

7.1. Overview

The 15-algorithm portfolio spans four orders of magnitude in runtime: FLIP on NH completes in 0.3 s; SIMANNEAL on TX requires 82.1 s. For a 50-state production sweep using the recommended configurations, practitioners need a compute budget estimate and a parallelism strategy. This section provides both.

7.2. Per-State Runtime Estimates

Table 11 gives estimated wall-clock runtime per state for the four most common production configurations. Estimates are derived from the five benchmark states (NC, WI, TX, FL, NH) by linear interpolation in $k \cdot N$ (the dominant cost factor for METIS-class algorithms).

The default configuration (METIS + Convergence sweep) completes a 50-state congressional sweep in approximately 500 s (≈ 8 minutes) on a single thread, or ≈ 30 s with 16-thread Rayon parallelism. This is the recommended configuration for exploratory runs and parameter tuning.

7.3. Practical Algorithm Selection Under Budget Constraints

Tight budget (< 5 minutes, single thread). Use METIS + Convergence (default) or BFS + ConvergenceSweep. ILP is applicable only to NH; exclude from production sweep or apply selectively. Do not use SA, CVD-Geo, PT, or SMC-Percentile.

Moderate budget (< 30 minutes, single thread or < 5 minutes with 16 threads). Use BFS or CVD-Geo structure + SMC-Percentile or Merge-Split search. This configuration covers PP-maximising and calibrated-distribution use cases at approximately $2\times$ the default runtime.

Full budget (< 2 hours, 16 threads). SA structure + PT search for EC-optimal results. CVD-Geo structure + SMC-Percentile search for PP + calibrated results. Adaptive Multi-scale for large- k states (TX, FL, CA, NY) for mixing quality.

7.4. Parallelism Strategy

The three-layer compositor exposes two levels of parallelism:

State-level parallelism. Each state is independent. For a 50-state sweep, distribute states across nodes with `-workers 1` per state and ≤ 50 parallel jobs. This is the most efficient parallelism strategy for homogeneous clusters.

Algorithm-level parallelism (ParTemp only). PARTEMP parallelises internally across replicas. With 8 replicas and `-workers 8` or `-workers 16`, PT achieves near-linear speedup up to 8 cores. For a node with 16 cores handling TX (the bottleneck state in any sweep), run 2 states in parallel each with 8 workers: `-workers 8 -replicas 8`.

Anti-pattern to avoid. Do not parallelise chain steps within a single state (e.g., using multiple threads for one Merge-Split chain). The Markov chain is inherently sequential; multi-threading a single chain provides no benefit and introduces synchronisation overhead. Only PARTEMP benefits from within-state multi-threading (replica parallelism).

7.5. Which Algorithms Are Practical for 50-State Sweeps

Table 12 classifies all 15 algorithms by production practicality for a 50-state congressional sweep.

7.6. Recommended 50-State Production Stack

For a complete 50-state congressional sweep that balances quality and runtime:

1. **Default pass:** All 50 states with `-structure prime-factor -weights county -search convergence`. Approximately 8 minutes single-threaded. This produces the official plan for all states.
2. **Large- k pass:** TX, FL, CA, NY, PA, OH ($k \geq 15$) with `-structure sa-bisect -search adaptive-multiscale -workers 16`. Approximately 40 minutes. Improves EC and ACF for the six states that most benefit from large- k methods.
3. **ILP pass:** NH only with `-structure ilp -search single`. Approximately 13 seconds. Certifies optimal EC for the two-district state.

4. **VRA pass:** Jurisdictions with active Section 2 claims with `-structure ratio-optimal -weights vra-aligned -search vra-recom`. Approximately 10–20 minutes depending on jurisdiction count.

Total estimated runtime for the complete four-pass stack on a 16-core machine: approximately 60–90 minutes.

8. Conclusion

8.1. Summary

We have compared all 15 implemented redistricting algorithms across both structure and search layers on five states (NC $k = 14$, WI $k = 8$, TX $k = 38$, FL $k = 28$, NH $k = 2$) and five metrics (EC, PP, D-seat, ACF(100), runtime). The main findings:

1. **ILP certifies what heuristics approximate.** For NH ($k = 2$), ILP achieves EC = 47 (certified optimal), 8.5% below Simulated Annealing (51) and 23.4% below METIS (58). The existence of the ILP baseline quantifies the optimality gap of every other structure algorithm for the first time.
2. **The structure layer determines the PP ceiling.** CVD-Geographic achieves the highest mean PP of any algorithm across all states, outperforming even SMC-Percentile (the best search method for PP). Search algorithms improve PP by at most 8% from the structure baseline; a poor PP structure choice cannot be fully compensated by search. When PP is the primary legal criterion, CVD-Geographic structure should be selected first, and the search algorithm selected second.
3. **Parallel Tempering dominates EC for large k .** For TX ($k = 38$) and FL ($k = 28$), PT achieves lower final EC than Short-Burst by escaping basin boundaries inaccessible to local burst strategies. The crossover from Short-Burst to PT is at approximately $k \approx 20$; for smaller k , Short-Burst is sufficient and faster on single-core hardware.
4. **Adaptive Multi-scale is the fastest-mixing chain for TX.** AMS reduces ACF(100) to 0.31 for TX vs. 0.37 for fixed Multi-scale, a 16% improvement, by front-loading coarse steps during warm-up. For $k < 15$, AMS and fixed Multi-scale are indistinguishable.

5. **The full portfolio is covered by six questions.** Two structure questions (certified optimality? PP priority?) and four search questions (VRA? $k \geq 15$? calibrated distribution? compactness optimisation?) route any redistricting task to the correct structure + search combination.

8.2. The Three-Layer Compositor as a Portfolio System

The three-layer compositor (`-structure`, `-weights-override`, `-search`) was designed to make algorithms composable. The comparison in this paper confirms that composability is not merely a software design property but a substantive algorithmic advantage: the best EC results require both a good structure algorithm (SA) and a good search algorithm (PT); the best PP results require both a good structure algorithm (CVD-Geo) and a calibrated search algorithm (SMC-Percentile). Neither layer alone achieves the best of both metrics simultaneously, and no single combination dominates all metrics.

This composability is the defining property of the platform. Prior redistricting tools typically implement one algorithm end-to-end (GerryChain’s ReCom, PlanScore’s baseline, redist’s SMC). The three-layer compositor turns algorithm selection into a parameter choice, enabling practitioners to express their redistricting requirements directly in the CLI and obtain the appropriate algorithm portfolio automatically.

8.3. Future Work

Confirmed interactions. The structure–search interaction (Section 6) is the primary open empirical question: does SA + PT compound (both improving EC), or does SA’s baseline already reduce PT’s marginal gain? A controlled experiment varying structure with fixed PT search, and varying search with fixed SA structure, would resolve this.

Additional structure algorithms. B.25 (in planning) will introduce a spectral bisection structure algorithm based on the Fiedler vector of the graph Laplacian. Spectral bisection is theoretically related to minimum-EC bisection and may bridge the gap between METIS and ILP for moderate- N instances.

Multi-run confidence intervals. All results in this paper are single-run. A follow-up study with $R = 20$ independent runs per algorithm–state pair would provide confidence

intervals and enable formal statistical comparison of the close results (e.g., PT vs. AMS for ACF; CVD-Geo vs. SMC-Percentile for PP).

G.15 as practitioner reference. This paper is intended as the definitive practitioner reference for the `redist` algorithm portfolio. As new algorithms are added to the B/G/H-series papers, they will be incorporated into the combined Table 9 and the six-question decision tree. The framework’s modular structure is designed to accommodate these additions without restructuring the existing decision logic.

8.4. Implementation Reference

All 15 algorithms are implemented in the `redist` Rust workspace:

- `crates/redist-core/src/structure/` — structure algorithms (`metis.rs`, `simulated_annealing.rs`, `cvd_graph.rs`, `cvd_geographic.rs`, `bfs_growth.rs`)
- `crates/redist-ilp/` — ILP redistricting crate (B.24)
- `crates/redist-ensemble/src/` — search algorithms (`shortburst.rs`, `flip.rs`, `forest_recom.rs`, `merge_split.rs`, `multiscale.rs`, `shortburst_forest.rs`, `vra_recom.rs`, `parallel_tempering.rs`, `smc.rs`, `smc_percentile.rs`, `adaptive_multiscale.rs`)
- `crates/redist-cli/src/compositor/` — three-layer compositor and CLI flag routing

The decision framework in Table 9 is the recommended reference for algorithm selection in the `redist` platform. All runs in this paper are reproducible from seed 42 and the configurations in Section 2.

References

Sarah Cannon, Ari Goldbloom-Helzner, Varun Gupta, J. N. Cory Matthews, and Bhushan Suwal. Voting rights, markov chains, and optimization by short bursts. *arXiv*, 2022. arXiv:2011.02322. Theoretical analysis of Short-Burst optimisation for redistricting Markov chains. Proves Short-Burst finds compact plans more efficiently than single-step gradient methods. Does not include PT, SMC, or multi-scale comparisons.

Gio Della-Libera. Adaptive multi-scale MCMC: Self-tuning coarse fraction for large- k redistricting, 2026a. B.21 in Algorithmic Redistricting series. Extends Multi-scale MCMC (G.11) with real-time adaptation of the coarse fraction α . Monitors coarse acceptance rate and adjusts α upward during warm-up. Reduces $\text{ACF}(100)$ to 0.31 for TX vs 0.37 for fixed Multi-scale. Recommended over fixed Multi-scale for $k \geq 20$; indistinguishable for $k < 15$.

Gio Della-Libera. BFS growth bisection: Zero-hyperparameter geographic district construction, 2026b. B.23 in Algorithmic Redistricting series. Implements breadth-first search growth from k random seeds. Fastest structure method ($O(N \log N)$, zero hyperparameters), with PP competitive with CVD-Graph. Recommended when structure runtime is the bottleneck and moderate PP is required without centroid data.

Gio Della-Libera. Bisection ensemble: Local ReCom at bisection nodes, 2026c. H.1 in Algorithmic Redistricting series. BisectionEnsemble runs a short ReCom chain at each node of the bisection tree during plan construction, selecting the best plan from each local ensemble. Combines structure-time search with top-down bisection for a hybrid approach.

Gio Della-Libera. Convergence sweep: Multi-seed federal statute default for redistricting, 2026d. B.16 in Algorithmic Redistricting series. ConvergenceSweep runs multiple seeds and selects the plan that converges first. Implements the federal statute default (–search convergence, threshold $T=600$). Recommended for production sweeps requiring a single defensible plan.

Gio Della-Libera. Centroidal Voronoi diagram redistricting: Graph-distance bisection, 2026e. B.22 in Algorithmic Redistricting series. Implements CVD bisection using graph hop-distance as the metric. Achieves higher mean Polsby-Popper than METIS without requiring centroid data. Recommended when PP is important and centroid data is unavailable or the input is a non-standard graph.

Gio Della-Libera. Geographic CVD redistricting: Euclidean-distance centroidal Voronoi bisection, 2026f. B.22b in Algorithmic Redistricting series. Phase 2 extension of B.22. Uses Euclidean distance from tract centroids rather than graph hop-distance, achieving the highest mean Polsby-Popper of any structure algorithm. Requires centroid data (INTPT-

LAT/INTPTLON fields). Recommended when PP is the primary structure criterion and centroid data is available.

Gio Della-Libera. Flip MCMC: Boundary-tract sensitivity analysis for redistricting, 2026g. G.8 in Algorithmic Redistricting series. Implements single-tract boundary flip Markov chain for local sensitivity analysis. High autocorrelation ($ACF(100)=0.83$ for TX) is intentional: characterises fragile boundary tracts at a fixed initial plan.

Gio Della-Libera. Forest ReCom: Reversible recombination via spanning forest sampling, 2026h. G.9 in Algorithmic Redistricting series. Implements ForestReCom with Metropolis-Hastings ratio. Provides approximately uniform distribution over valid plans. Baseline for approximately uniform chain comparisons.

Gio Della-Libera. A practitioner’s comparison of redistricting ensemble algorithms, 2026i. G.14 in Algorithmic Redistricting series. Companion paper to G.15. Compares eight search algorithms (Short-Burst, SMC, Flip, Forest ReCom, Merge-Split, Multi-scale, ShortBurstForest, VRA-Aware) on NC, WI, TX using four metrics. G.15 extends this to 15 algorithms, five states, and five metrics, and adds the structure-layer comparison.

Gio Della-Libera. ILP redistricting: Exact minimum-EC bisection for small instances, 2026j. B.24 in Algorithmic Redistricting series. Implements branch-and-bound ILP for certified-optimal minimum edge-cut bisection. Applicable when $N \leq 500$ per bisection half (e.g., NH $k=2$ with $N=260$ tracts). Achieves $EC = 47$ for NH (certified optimal), 23.4% below METIS and 8.5% below Simulated Annealing. Includes `-ilp-timeout` fallback to SA.

Gio Della-Libera. Merge-Split MCMC: Balanced recombination for redistricting, 2026k. G.10 in Algorithmic Redistricting series. Merge-Split chain with approximately uniform target distribution. Recommended as the general-purpose algorithm for small- k redistricting without special requirements. Marginally lower ACF than Forest ReCom.

Gio Della-Libera. Multi-scale MCMC: Hierarchical mixing for large- k redistricting, 2026l. G.11 in Algorithmic Redistricting series. Implements Multi-scale MCMC Option B (tract-county) using GEOID prefix coarsening. Recommended for $k \geq 15$; reduces $ACF(100)$ by 27–43% relative to single-scale chains. Adaptive Multi-scale (B.21) further improves ACF for TX.

Gio Della-Libera. Percentile sweep: Statutory choice of legal posture via percentile selection, 2026m. H.0 in Algorithmic Redistricting series. PercentileSweep runs multiple seeds and selects the plan at the p -th percentile of the EC distribution, enabling practitioners to express a legal posture (conservative, median, aggressive) directly. Extension of ConvergenceSweep with percentile parameter.

Gio Della-Libera. Parallel tempering for redistricting ensembles: Multi-basin search via replica exchange, 2026n. B.20 in Algorithmic Redistricting series. Implements replica exchange MCMC with 8 replicas across a temperature ladder. Achieves the lowest final EC among search algorithms for TX ($k=38$) and FL ($k=28$) by escaping compact-plan basins inaccessible to local burst strategies. Requires 8–16 cores for practical use; recommended for large- k EC optimisation.

Gio Della-Libera. Simulated annealing bisection for redistricting: EC-optimal heuristic structure, 2026o. B.19 in Algorithmic Redistricting series. Implements METIS-initialised simulated annealing on the bisection graph. Achieves the best heuristic EC among structure methods (10–11% below METIS); 12x slower than METIS baseline. Recommended when EC minimisation is the primary structure criterion and runtime is not constrained.

Gio Della-Libera. Short-Burst MCMC: Compactness optimisation via repeated exploration bursts, 2026p. G.6 in Algorithmic Redistricting series. Implements Short-Burst ReCom with burst length and restart schedule. Primary compactness optimisation algorithm for small- k states ($k < 20$). For $k \geq 20$, Parallel Tempering (B.20) achieves lower final EC by escaping inter-basin barriers.

Gio Della-Libera. ShortBurstForest: Compactness optimisation on forest ReCom chains, 2026q. G.12 in Algorithmic Redistricting series. Runs Short-Burst optimisation on the ForestReCom chain, combining compactness optimisation with MH-corrected distributional properties. Second-lowest EC for small- k ; recommended when distributional awareness is required alongside compactness.

Gio Della-Libera. SMC-Percentile: Calibrated plan selection via sequential Monte Carlo percentile extraction, 2026r. Extension of G.7 (SMC). SMC-Percentile uses the same particle cloud as standard SMC but returns the plan at the p -th percentile of the PP distribution among particles (default $p=50$, the median). Achieves the highest PP among all

search algorithms; recommended when the reported plan must be described as “the median plan under the calibrated distribution.” Implemented as `SeedCompositor::SmcPercentile` in `redist-cli`.

Gio Della-Libera. VRA-aware ReCom: Minority-majority district preservation via adaptive sampling, 2026s. G.13 in Algorithmic Redistricting series. VRA-constrained ReCom chain with adaptive boosts to maintain minority-majority districts throughout sampling. Only algorithm that enforces VRA constraints during chain execution. Required for VRA Section 2 analysis.

Svante Janson. Graphons, cut norm and distance, couplings and rearrangements. *arXiv*, 2022. arXiv:2108.08753. Graph-cut framework used to understand autocorrelation of the EC statistic in redistricting chains. Theoretical basis for the ACF comparisons in Sections 5 and 4.

George Karypis and Vipin Kumar. A fast and high quality multilevel scheme for partitioning irregular graphs. *SIAM Journal on Scientific Computing*, 20(1):359–392, 1998. doi: 10.1137/S1064827595287997. The METIS multilevel k-way partitioning algorithm. The structure-layer baseline in G.15. Fast and well-tested; the default `-structure standard-bisect` flag invokes the bundled METIS C library via `redist-metis FFI`.

Table 9: Combined structure + search recommendations by use case. “Alternative” is the second choice when the primary is unavailable. All flags are for the `redist` CLI.

Use Case	Recommended	Alternative
Court submission (calibrated, neutral)	<code>-structure</code> <code>prime-factor</code> <code>-search</code> <code>smc-percentile</code>	<code>-structure</code> <code>prime-factor</code> <code>-search merge-split</code>
EC minimisation, small state ($k \leq 4$, $N \leq 500$)	<code>-structure ilp</code> <code>-search single</code>	<code>-structure sa-bisect</code> <code>-search short-burst</code>
EC minimisation, large state	<code>-structure sa-bisect</code> <code>-search</code> <code>parallel-tempering</code>	<code>-structure sa-bisect</code> <code>-search short-burst</code>
PP maximisation	<code>-structure</code> <code>cvd-geographic</code> <code>-search</code> <code>smc-percentile</code>	<code>-structure</code> <code>bfs-growth</code> <code>-search merge-split</code>
Large- k ($k \geq 15$), fast mixing	<code>-structure</code> <code>prime-factor</code> <code>-search</code> <code>adaptive-multiscale</code>	<code>-structure</code> <code>prime-factor</code> <code>-search multiscale</code>
Large- k , EC optimisation (multi-core available)	<code>-structure sa-bisect</code> <code>-search</code> <code>parallel-tempering</code>	<code>-structure</code> <code>standard-bisect</code> <code>-search short-burst</code>
VRA Section 2 analysis	<code>-structure</code> <code>ratio-optimal</code> <code>-weights vra-aligned</code> <code>-search vra-recom</code>	—
Local sensitivity at a fixed plan	<code>-structure</code> <code>ratio-optimal</code> <code>-search flip</code>	—
Production sweep, general purpose	<code>-structure</code> <code>prime-factor</code> <code>-weights county</code> <code>-search convergence</code>	<code>-structure</code> <code>prime-factor</code> <code>-search merge-split</code>
Fastest result (time budget limited)	<code>-structure</code> <code>bfs-growth</code> <code>-search convergence</code>	<code>-structure</code> <code>standard-bisect</code> <code>-search single</code>

Table 10: Algorithm sensitivity by metric. Layer: which layer (Structure, Search, or Both) most determines the metric. Leader: the algorithm achieving the best value on this metric in this comparison.

Metric	Layer	Leader	Second
EC (final plan)	Both	PARTEMP + SIMANNEAL	SHORTBURST + SIMANNEAL
PP (final plan)	Structure (primary)	CVD-GEO + SMC-PERCENTILE	CVD-GEO + SMC
D-seat	Search (modest)	Varies by state	—
ACF(100)	Search only	ADAPTIVEMS	PARTEMP
Runtime (total)	Both	BFSGROWTH + FLIP	METIS + FLIP

Table 11: Estimated wall-clock runtime per state (seconds) for four production configurations. Estimates for k, N values not in the benchmark states are interpolated. “16w” indicates **-workers 16**; all others **-workers 1**. These are rough estimates ($\pm 30\%$) for planning purposes.

Configuration	NC ($k = 14$)	WI ($k = 8$)	TX ($k = 38$)	FL ($k = 28$)	Typical
METIS + Convergence (default)	~ 12	~ 5	~ 38	~ 28	
SA + Short-Burst (EC optimal)	~ 40	~ 18	~ 135	~ 98	
CVD-Geo + SMC-Percentile (PP + calibrated)	~ 27	~ 12	~ 95	~ 72	
SA + PT-16w (EC + large- k optimal)	~ 37	~ 16	~ 124	~ 90	

Table 12: Production practicality for a 50-state congressional sweep. “Full sweep” means all 50 states; “Select” means large- k states only or NH only; “Not practical” means runtime or hardware requirements preclude routine use.

Algorithm	Production Practicality	Notes
METIS	Full sweep	Default; fastest
BFS Growth	Full sweep	Slightly faster than METIS
CVD-Graph	Full sweep	$2\times$ METIS runtime
CVD-Geo	Full sweep	Requires centroid data; $2\times$ METIS
SA	Full sweep	$10\times$ METIS; budget ~ 30 min/sweep
ILP	NH only	Applicable to $k = 2$ states; $60\times$ METIS
ConvergenceSweep	Full sweep	Default search; fast
PercentileSweep	Full sweep	$\sim 1.5\times$ Convergence
BisectionEnsemble	Full sweep	$\sim 2\times$ Convergence
Merge-Split	Full sweep	Recommended general-purpose
Forest ReCom	Full sweep	Similar to Merge-Split
Short-Burst	Full sweep	EC optimal, single-core
ShortBurstForest	Full sweep	EC + MH; $\sim 50\%$ slower than Short-Burst
SMC	Full sweep	Calibrated; $\sim 5\times$ Merge-Split
SMC-Percentile	Full sweep	Calibrated point estimate; $\sim$ SMC cost
VRA-Aware	Select	VRA jurisdictions only
Flip	Select	Sensitivity analysis at a fixed plan
Multi-scale	Large- k	TX, FL, CA, NY, PA, OH; not needed for $k < 15$
Adaptive MS	Large- k	Preferred over fixed Multi-scale for $k \geq 20$
Parallel Tempering	Large- k (multi-core)	8–16 cores; EC optimal for large- k